

Labwork 4

HPC – 2D grid for grayscale

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1 Preparation

The image is the same as in labwork 3, `astronaut()` from `skimage.data`, size 512×512 with 3 channels. In labwork 3 I flattened it to $(262144, 3)$ and used a 1D grid. Here the image keeps its shape $(512, 512, 3)$ and the threads are organised in a 2D grid, so a thread finds its pixel with a row index and a column index instead of a single index.

2 Results

I used the same method as in labwork 3: the image stays on the device so only the kernel is measured, each configuration is launched once as a warm-up and then 100 times, and I take the average.

Table 1: Kernel time for the 2D block sizes (Tesla T4 on Google Colab).

Block size	Threads	Grid	Time (ms)
(8, 8)	64	(64, 64)	0.0993
(16, 8)	128	(32, 64)	0.0469
(16, 16)	256	(32, 32)	0.0411
(32, 8)	256	(16, 64)	0.0402
(16, 32)	512	(32, 16)	0.0375
(32, 32)	1024	(16, 16)	0.0376
(64, 16)	1024	(8, 32)	0.0404

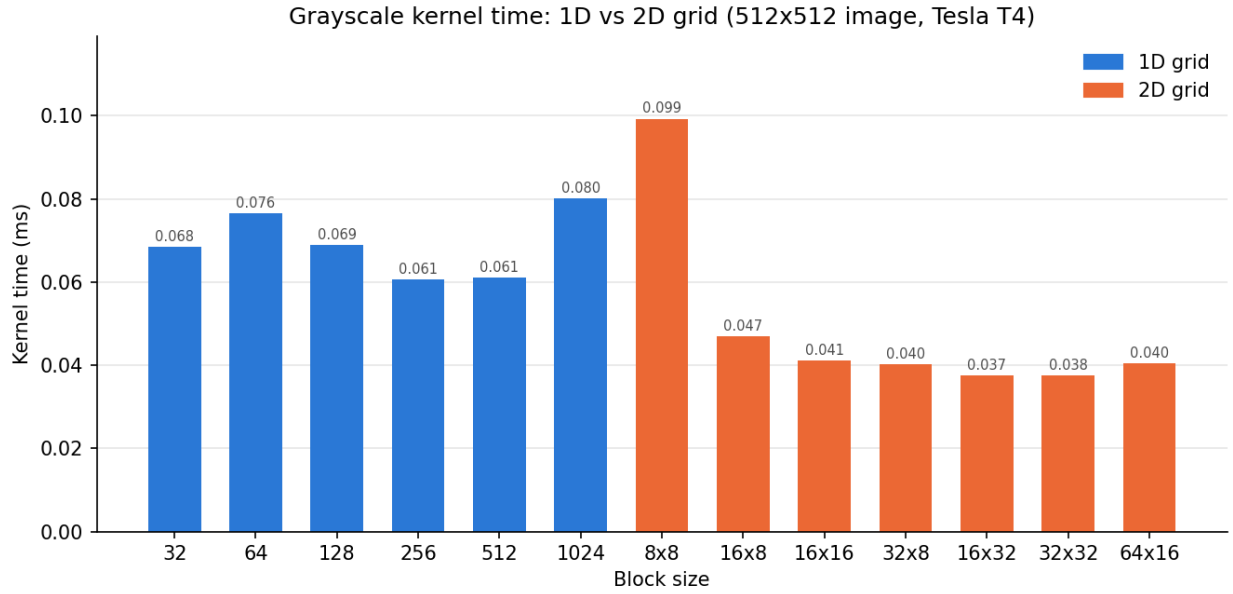


Figure 1: Kernel time of the 1D version (blue) and the 2D version (orange).

The best 2D configuration is (16, 32) with 0.0375 ms and the best 1D one is 256 threads and 512 threads with 0.0607 ms.

3 Comment

The 2D grid is more convenient to write, do not have to reshape anything.